

ARTIFICIAL INTELLIGENCE

ASSESSMENT TOOL 1

Knowledge Representation and Logical Reasoning

Course Name	Artificial Intelligence
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CASE STUDY 1 – SMART MEDICAL DIAGNOSIS SYSTEM

(a) Propositional Logic Representation

Let:

- F = Patient has Fever
- C = Patient has Cough
- R = Patient has Rash
- B = Patient has Breathlessness
- Flu = Possible Flu
- Measles = Possible Measles
- Pneumonia = Possible Pneumonia

Rules

1. Fever AND Cough $\rightarrow$ Possible Flu

$$F \wedge C \rightarrow \text{Flu}$$

2. Fever AND Rash $\rightarrow$ Possible Measles

$$F \wedge R \rightarrow \text{Measles}$$

3. Cough AND Breathlessness $\rightarrow$ Possible Pneumonia

$$C \wedge B \rightarrow \text{Pneumonia}$$

Facts of Patient A

F = True

C = True

B = True

R is not given.

(b) Forward Chaining

Forward Chaining starts with the known facts and applies rules to derive new facts.

Step 1

Facts: F = True, C = True, B = True

Rule 1: $F \wedge C \rightarrow \text{Flu}$

Since Fever and Cough are true: Flu = True

Step 2

Rule 2: $F \wedge R \rightarrow \text{Measles}$

Rash is not given.

Therefore: Measles cannot be derived.

Step 3

Rule 3: $C \wedge B \rightarrow \text{Pneumonia}$

Cough and Breathlessness are true.

Therefore: Pneumonia = True

Final Conclusions

- Possible Flu = True
- Possible Pneumonia = True
- Possible Measles = Cannot be concluded

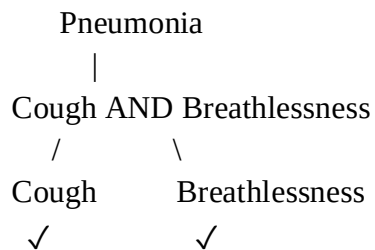
(c) Backward Chaining for Pneumonia

Goal: Pneumonia

To prove Pneumonia, find a rule whose conclusion is Pneumonia.

$C \wedge B \rightarrow \text{Pneumonia}$

Therefore, subgoals are: Cough AND Breathlessness



Both facts are available for Patient A.

Therefore: Pneumonia is proved.

CASE STUDY 2 – AUTONOMOUS TRAFFIC MANAGEMENT AGENT

The case uses First-Order Logic (FOL).

(a) Unification

Rule 1

$\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Given facts:

- $\text{Vehicle}(\text{Ambulance})$
- $\text{EmergencyType}(\text{Ambulance})$

We substitute: $x = \text{Ambulance}$

Therefore the substitution set is: $\theta = \{x / \text{Ambulance}\}$

After substitution: $\text{Vehicle}(\text{Ambulance}) \wedge \text{EmergencyType}(\text{Ambulance})$

Both facts are true.

Therefore: $\text{ClearPath}(\text{Ambulance})$

MGU: $\theta = \{x \rightarrow \text{Ambulance}\}$

(b) Resolution Method

Goal: $\text{Proceed}(\text{CarA})$

Relevant facts:

1. $\text{Vehicle}(\text{CarA})$
2. $\text{Behind}(\text{CarA}, \text{Ambulance})$
3. $\text{Vehicle}(\text{Ambulance})$
4. $\text{EmergencyType}(\text{Ambulance})$

Rule 1

$\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

CNF: $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Using $\text{Vehicle}(\text{Ambulance})$ and $\text{EmergencyType}(\text{Ambulance})$, we derive:

$\text{ClearPath}(\text{Ambulance})$

Rule 2

$\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$

Therefore: $\text{GreenSignal}(\text{Ambulance})$

Rule 3

$\text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$

Substitute: $x = \text{CarA}$, $y = \text{Ambulance}$

- $\text{Vehicle}(\text{CarA}) \checkmark$
- $\text{Behind}(\text{CarA}, \text{Ambulance}) \checkmark$
- $\text{GreenSignal}(\text{Ambulance}) \checkmark$

Therefore: $\text{Proceed}(\text{CarA})$

Final Result: $\text{Proceed}(\text{CarA})$ is TRUE.

(c) Two Simultaneous Emergency Vehicles

The current knowledge base can handle individual emergency vehicles because the rules use $\forall x$.

However, to properly handle two simultaneous emergency vehicles, the KB should contain:

5. Facts identifying both vehicles as emergency vehicles.
6. Their positions or routes.
7. Rules for prioritizing emergency vehicles.
8. Rules for handling conflicts between two emergency routes.
9. Rules for preventing collision.
10. Separate signal control for each affected road.

Conclusion: The current KB provides basic emergency routing, but additional conflict-resolution and route-management rules are required for simultaneous emergencies.

CASE STUDY 3 – AGRICULTURAL EXPERT REASONING SYSTEM

The given knowledge base (KB) is:

- $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- $\text{SoilDry} = \text{True}$
- $\text{CropWheat} = \text{True}$

(a) Convert to CNF

P1

$\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

Using: $P \rightarrow Q \equiv \neg P \vee Q$

Therefore: $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

P2

$\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

Convert: $\neg(\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$

Using De Morgan's law:

$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

P3

$\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Therefore: $\neg(\neg \text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$

Using De Morgan's law:

$\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

Facts

SoilDry

CropWheat

So the CNF clauses are:

1. $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$
2. $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$
3. $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$
4. SoilDry
5. CropWheat

(b) Resolution Refutation

Goal

ApplyDripMethod

Negate the goal: $\neg \text{ApplyDripMethod}$

Add it to the KB.

We have: SoilDry

From clause 1: $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

Resolution gives: IrrigationNeeded

Now we have: IrrigationNeeded and CropWheat

Using clause 2: $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

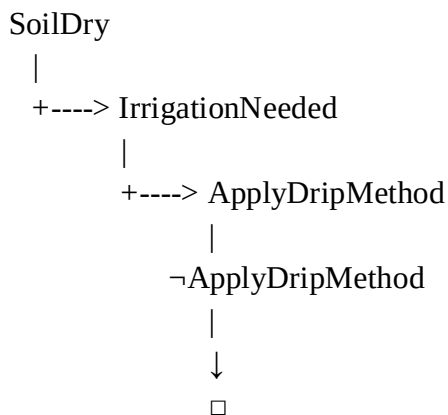
Resolution gives: ApplyDripMethod

But we also assumed: $\neg \text{ApplyDripMethod}$

Therefore: $\text{ApplyDripMethod} \wedge \neg \text{ApplyDripMethod}$

This produces the empty clause $\square$.

Resolution Tree



Final Result: ApplyDripMethod is proved.

(c) Remove SoilDry

Suppose: SoilDry is removed.

Then Rule 1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$ cannot be triggered.

Therefore: IrrigationNeeded cannot be derived.

Without IrrigationNeeded, Rule 2 cannot derive: ApplyDripMethod

However, CropAtRisk requires: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat}$

The KB does not explicitly provide $\neg \text{ApplyDripMethod}$.

Therefore: CropAtRisk cannot be logically concluded from the remaining KB.

Final Conclusion

- IrrigationNeeded $\rightarrow$ Not derivable
- ApplyDripMethod $\rightarrow$ Not derivable
- CropAtRisk $\rightarrow$ Not derivable

CASE STUDY 4 – WUMPUS WORLD KNOWLEDGE AGENT

The given percepts are:

- Stench at [1,2]
- Breeze at [1,1]
- Glitter at [2,2]

Rules are given for Wumpus, Pit, Gold and Safe states.

(a) Belief State and Modus Ponens

Percepts

Stench[1,2]

Breeze[1,1]

Glitter[2,2]

Rule 1

Stench[1,2] $\rightarrow$ Wumpus is adjacent to [1,2]

Since Stench[1,2] is true: Wumpus is adjacent to [1,2]

Rule 2

Breeze[1,1] $\rightarrow$ Pit is adjacent to [1,1]

Since Breeze[1,1] is true: Pit is adjacent to [1,1]

Rule 3

Glitter[x,y] $\rightarrow$ Gold[x,y]

Given Glitter[2,2], therefore: Gold[2,2]

Rule 4

$\neg$ Wumpus[x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

A cell can be declared safe only when both absence conditions are known.

Derived Knowledge

Stench[1,2]

↓

Wumpus adjacent to [1,2]

Breeze[1,1]

↓

Pit adjacent to [1,1]

Glitter[2,2]



Gold[2,2]

(b) Forward Chaining

Step 1

Percept: Stench[1,2]

Therefore: Wumpus is adjacent to [1,2]

Step 2

Percept: Breeze[1,1]

Therefore: Pit is adjacent to [1,1]

Step 3

Percept: Glitter[2,2]

Therefore: Gold[2,2]

Possible Locations

- Wumpus is adjacent to [1,2].
- Pit is adjacent to [1,1].
- Gold is at [2,2].

The rules provided do not specify exact neighboring cells, so an exact single location for Wumpus or Pit cannot be determined from the given KB alone.

(c) Safety of Path

Path: [1,1] → [1,2] → [2,2]

Cell [1,1]

There is a Breeze at [1,1].

Therefore: Pit is adjacent to [1,1].

So the path cannot be declared completely safe based only on the provided knowledge.

Cell [1,2]

There is a Stench at [1,2].

Therefore: Wumpus is adjacent to [1,2].

Again, the exact location of Wumpus is not known.

Cell [2,2]

There is Glitter at [2,2].

Therefore: Gold[2,2]

Final Conclusion

The agent cannot guarantee that the complete path is safe using the given KB.

The agent knows that gold is at [2,2], but the presence of Breeze and Stench indicates possible hazards around the path.

Therefore, the path should not be considered fully safe without additional knowledge.

FINAL ANSWERS AT A GLANCE

Case Study	Main Result
1. Medical Diagnosis	Flu = True, Pneumonia = True
2. Traffic Management	Proceed(CarA) = True
3. Agricultural System	ApplyDripMethod = True
4. Wumpus World	Gold at [2,2]; path safety cannot be guaranteed